Karlsruhe Institute of Technology
Institute of Control Systems
Chair for Secure Interconnected Automation Technology
Univ.-Prof. Dr.-Ing. Mike Barth

Unified Digital Twin Framework for Human, Prosthetic and Robotic Finger Dynamics

Unified Physics-Based Finger Model Using OpenModelica

Author: Athul Sunikumar, Bergin Sucil Dhas, Cris Cyrus

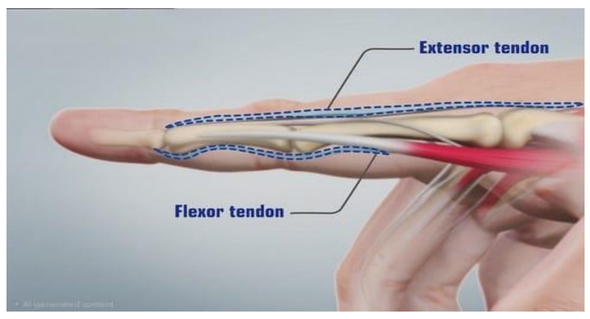

Figure 1: Finger Anatomy

Introduction & Motivation

Motivation:

Physical prototyping of finger systems for prosthetics, robotics, or research is costly and slow. Engineers need tools to explore design alternatives before building hardware.

Approach:

FingerJointTwin enables virtual experimentation. Test different materials, actuation strategies, and control algorithms in simulation. Explore hundreds of design variations in hours instead of months. Validate concepts digitally before committing to physical prototypes. The framework serves prosthetic designers, roboticists, and biomechanics researchers, accelerating innovation across all three domains.

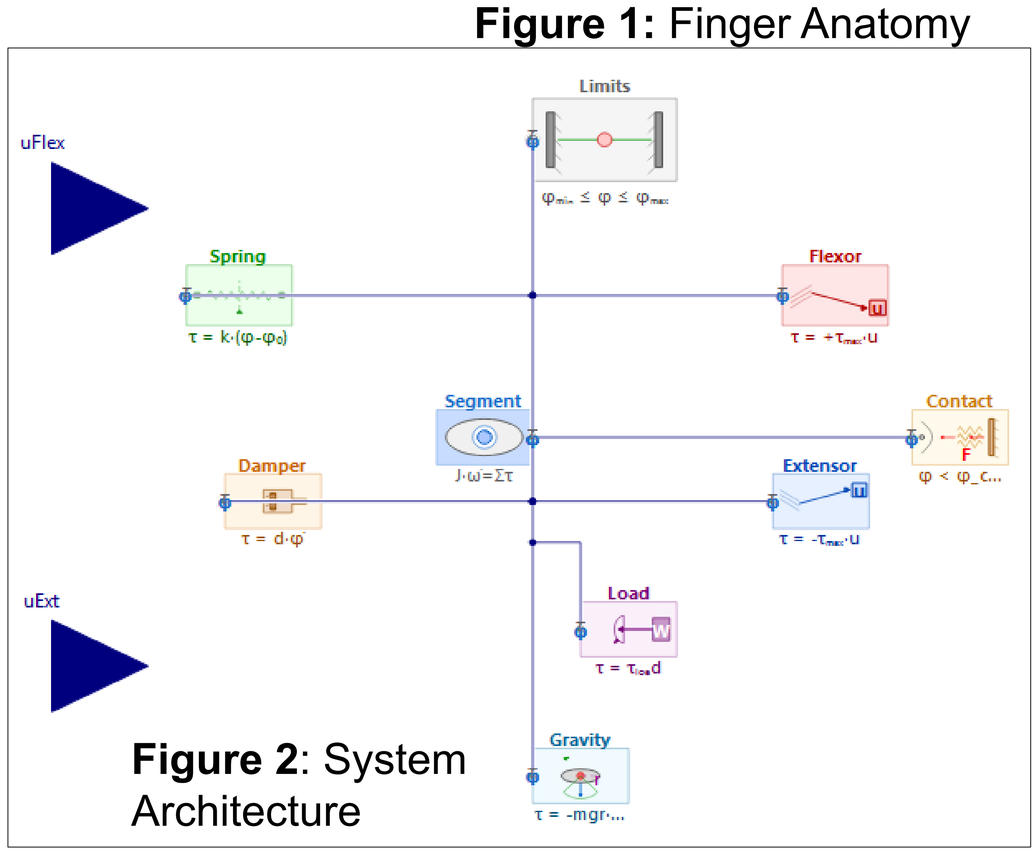

Figure 2: System Architecture

How the Digital Twin Works

🎯 **Modular Design:**

All components connect to one rotating joint
Each contributes torque independently
Components:

- FlexorTendon & ExtensorTendon (actuation)
- JointSpring, JointDamper, JointLimits (passive tissue)
- GravityTorque, ExternalLoad, ContactLoad (environment)
- FingerSegment

💪 **What Makes It Move:**

- Muscles pull finger closed/open ($\tau = \tau_max \cdot u$)
- Spring pulls back to neutral ($\tau = k \cdot \varphi$)
- Damper slows motion ($\tau = d \cdot \dot{\varphi}$)
- Object pushes back when touched (contact model)

⚙️ **Physics Engine:**

$J \cdot \dot{\omega} = \Sigma \tau_i$ (Newton's law for rotation)
Where: J = inertia, ω = angular velocity, τ_i = component torques
Same framework, different parameters →Human (gentle) to Robot (powerful) grasping

References:
[1] Brook, N. et al. (1995). A biomechanical model of index finger dynamics. Medical Engineering & Physics, 17(1).
[2] Machado, M. et al. (2012). Compliant contact force models in multibody dynamics. Mechanism and Machine Theory, 53.
[3] Belter, J.T. et al. (2013). Mechanical design of anthropomorphic prosthetic hands. Journal of Rehabilitation Research and Development, 50(5).

Simulation Results

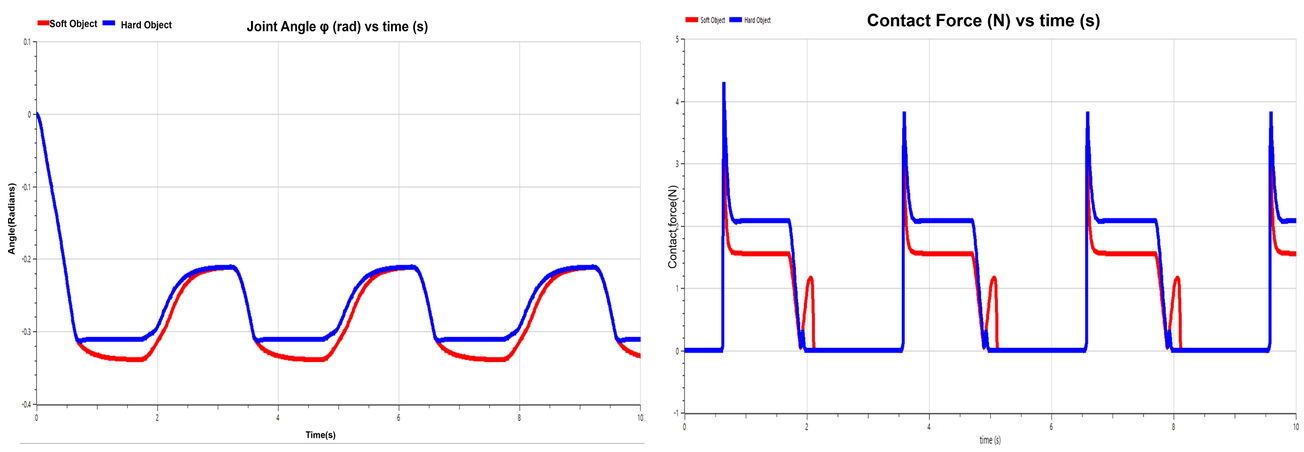

Figure 3: Object Stiffness Effect - Case 2 (Soft Object) vs Case 3 (Hard Object)

Results & Applications

Validated Across Three Domains:

Human Finger:

- Natural grasping behaviour with soft tissue compliance
- Torque-control validated: stiffness affects penetration > force
- Cases 2 vs 3: 5× stiffer object → 40% less penetration, only 16% more force
- muscle activation limits force, not object stiffness

Prosthetic Finger:

- Design parameter exploration without hardware prototyping
- Workspace constraints identified (stiff joint limits motion)
- Enables optimization before manufacturing

Robotic Finger:

- High-performance grasping with fast controlled actuation
- Contact force: ~ 40 N
- Demonstrates framework scales across performance range

Conclusion:

1. Key Finding: Same physics, different parameters → Human (gentle) to Robot (powerful) grasping
2. Framework enables design exploration without hardware prototyping

Athul Sunilkumar: ushet@student.kit.edu
Bergin Sucil Dhas: utwla@student.kit.edu
Cris Cyrus: uqhvj@student.kit.edu

Institut für Regelungs- und Steuerungstechnik
Fritz-Haber-Weg 1, 76131 Karlsruhe
Geb. 30.33